predict the risk of heart attacks. This AI solution can pinpoint new risk factors and assign a heart risk score to patients without comprehensive health check-ups, thus enabling early identification. AI-driven portable screening devices broaden the scope of eye screenings to prevent blindness and enhance accessibility in remote areas. Through AI, cardiovascular event prediction now hinges on analysing an individual's eye scan, potentially replacing traditional methods like CT scans, MRIs and X-rays.

AI holds the promise of developing novel, less invasive heart health tests. It not only identifies vulnerable patients early but also aids in tailoring treatment intensity by categorising them into high-risk or very high-risk groups. AI algorithms enable doctors to analyse a broader array of data and predict, with increased precision, personalised drug combinations.

Leveraging cloud capabilities, doctors can share high-resolution 3D imagery and consult local and international partners in real time, improving pre-planning and the overall procedural success. AI also enables access to learning and data from numerous patient cases, providing physicians with a comprehensive overview of potential diagnoses. Moreover, the technology's promise extends to supporting young physicians too, helping them reach accurate conclusions even with limited experience. Given such positives, AI literacy has to be introduced to clinicians so that it's treated as any other tool at their disposal.

The Affordability Gap

In 1991, India reached a record doctor-to-patient ratio of 1.2 doctors per 1,000 patients, as per the World Health Organization (WHO). However, due to population growth, this ratio dropped to around 0.7 by 2020. In rural areas in particular, the scarcity of specialised care is notably severe. As per government records from March 2022, there was around 80 per cent shortage of surgeons, physicians, gynaecologists and paediatricians at community health centres in rural India. The count of 4,485 specialists fell far short of the required 21,920, as stated by government data. Affordability continues to pose a challenge, with nearly 70 per cent of healthcare expenses being privately funded—among the highest globally. This disproportionately affects the poor, leading to an estimated 63 million people facing poverty annually due to healthcare costs.

AI presents an opportunity to render healthcare services more affordable and accessible, especially in remote regions. For instance, currently, continuous monitoring is limited to ICU beds, which account for only around 100,000 of nearly 2 million beds nationwide. AI can extend monitoring to any bed at a tenth of the cost of a regular ICU bed. In IP (in-patient) settings, with AI-driven models, clinicians can focus on high-risk patients by analysing symptoms as part of the Early Warning System, and can help in reducing mortality rates and average length of stay in hospitals.

As an administrative tool, AI can contribute to time-saving by automating routine clinical tasks, freeing up healthcare professionals to concentrate on crucial decision-making. Chatbots, for instance, assist patients in handling inquiries related to appointments, billing and other matters. Virtual health assistants manage patient queries through calls and emails, arrange doctor appointments, issue follow-ups and send clinical appointment reminders. Embedded with AI, remote patient monitoring systems facilitate proactive interventions and personalised care for individuals with chronic conditions.

All in all, AI-powered healthcare acts as an informative aide, helping doctors in extracting meaningful patterns from collected data. This can potentially save considerable time and effort by providing easy access to unbiased, consistent, high-quality diagnoses and treatments. The adoption of AI tools, like deep learning or machine learning, in routine medical decision-making is set to enhance patient care and affordability. It can be a transformative leap forward. As AI continues to evolve, it stands poised to revolutionise healthcare delivery, making predictable disease, precise medicine and customisable treatment and thus, ultimately, improving patient outcomes. However, all this data, which could sometimes even cover entire populations, must be guarded against breaches and fortified for quality. Any inaccuracy could have devastating consequences for patients. Embracing this technology with cautious optimism promises to reshape the future of medicine and healthcare, offering innovative solutions to age-old challenges and paving the way for a healthier society. ■

TAKEAWAYS

- With the myriad data it collects, AI can identify risks early and help in timely treatment
- AI can automate routine tasks, thus freeing up doctors for more crucial decision-making
- However, any breach or inaccuracy in data can be devastating for patients